gladly tackle difficult or dangerous assignments like jumping into black holes. They're true geek superheroes!

Let's say that Alice and Bob have provided two systems—Alice's system and Bob's system. Alice's system—whatever it is—is described by a space of states called S_A, and similarly Bob's system is described by a space of states called S_B.

Now let's say that we want to combine the two systems into a single composite system. Before going any further, let's be more specific about the systems we're starting with. For example, Alice's system could be a quantum mechanical coin with two basis states *H* and *T*. Of course, a classical coin must be in either one state or the other, but a quantum coin can exist in a superposition:

$$\alpha_H|H\} + \alpha_T|T\}.$$

You'll notice that I've used an unusual notation for Alice's ket-vectors. This is to distinguish them from Bob's kets. The new notation is intended to discourage us from adding vectors in Alice's space S_A to vectors in Bob's space S_B. Alice's S_A is a two-dimensional vector space—it is defined by the two basis vectors $|H\}$ and $|T\}$.

Bob's system might also be a coin, but then again it might be something else. Let's assume it's a quantum die. Bob's space of states S_B would then be six-dimensional, with the basis